Supplementary information for: A gas-transport model predicts the *Arabidopsis* spaceflight transcriptome, and hardware ventilation resolves true microgravity responses from sealed canister artifacts

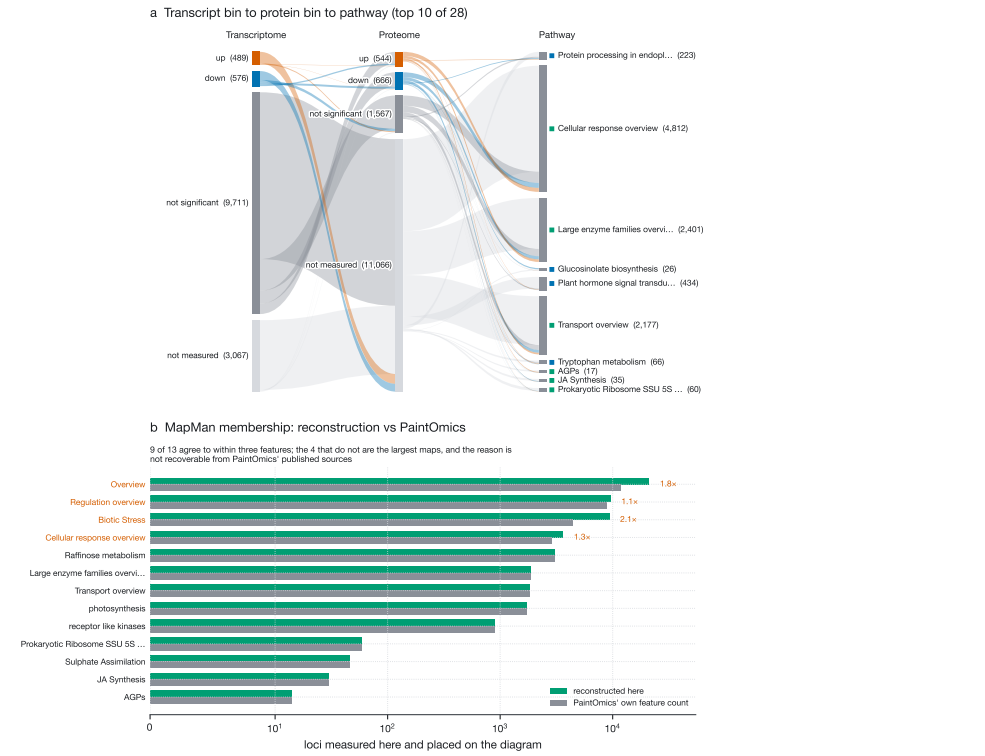
Richard Barker[1*]

[1]Department of Botany, University of Wisconsin-Madison, Madison, WI, USA.

Corresponding author(s). E-mail(s): dr.richard.barker@gmail.com;

Supplementary Figures and Tables

Supplementary Figure S1 — transcript, protein and pathway together



Ribbon width is a locus count. Columns 1 and 2 conserve loci exactly; column 3 counts memberships. Three whole-ontology MapMan maps are excluded (each covers more than 25% of measured loci). Blue square = KEGG, green = MapMan, as in Fig. 6a.

Fig. S1 Locus-level multi-omics flow and MapMan ontology reconstruction. a, Each locus in at least one of the 28 non-whole-ontology significant pathways is placed in one transcript bin and one protein bin, and then into the pathways it belongs to. Ribbon width is proportional to locus count. The pathway column draws the ten most significant pathways; a square gives each pathway's database in the colours of Fig. 6a. **b**, Reconstructed MapMan diagram membership compared against the feature count PaintOmics reported for each diagram (log scale). Nine of the thirteen diagrams agree to within three features.

Table S1 Cross-hardware pathway classification: comparison of the key pathways enriched in sealed BRIC-LED (OSD-522) against ventilated ABRS (OSD-7).

DB	Pathway	BRIC-LED p	ABRS p	ABRS Dir	Classification
K	Protein processing in ER	8.65×10^{-9}	0.9456	Flat	Hardware artifact (volatiles / CO_2)
M	Cellular response overview	2.33×10^{-6}	0.6111	Down	Hardware artifact
M	Biotic Stress	2.10×10^{-4}	0.0151	Up	**Conserved microgravity response**
K	Glucosinolate biosynthesis	6.02×10^{-4}	0.8453	Up	Hardware artifact
K	Plant hormone signal transduction	0.0011	0.0516	Up	Borderline conserved
K	Photosynthesis	0.0080	0.8512	Down	Hardware-modulated
K	MAPK signaling pathway	0.0094	0.2322	Up	Hardware-modulated
K	Starch and sucrose metabolism	0.0186	0.5428	Down	Hardware-modulated
K	Photosynthesis - antenna proteins	0.0217	1.0000	Down	Hardware-modulated
M	Raffinose metabolism	0.0274	0.0081	Up	**Conserved microgravity response**
K	Carbon fixation by Calvin cycle	0.1420	0.0074	Up	ABRS-specific (CO_2 saturated)
K	Cysteine and methionine metabolism	0.2110	0.0015	Down	ABRS-specific